

VIVEK GUPTA

Sunnyvale, CA (open to relocation) | +1-765-701-9946 | gupta96v@gmail.com | [linkedin.com/in/guptav96](https://www.linkedin.com/in/guptav96) | github.com/guptav96

PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience designing and scaling real-time, distributed backend systems, event-driven data pipelines, and ML-enabled decisioning infrastructure. Currently serves as technical lead on fraud-platform initiatives at Walmart Global Tech, contributing to a platform that prevents ~\$2.5B in annual fraud losses across billion-scale transaction volume.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript, C++, C#, Scala

Backend & Data: Kafka, REST APIs, GraphQL, Apache Spark, microservices, event-driven architecture, GraalVM, RabbitMQ, Nginx

Databases & Cloud: MariaDB, MongoDB, Cassandra, Spanner, BigQuery, MySQL, PostgreSQL, Redis, Elasticsearch, Neo4j, DynamoDB, GCP, AWS

DevOps & ML: Docker, Kubernetes, Jenkins, Grafana, Prometheus, PyTorch, TensorFlow, NumPy, Pandas, scikit-learn

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Fraud Platform

Jan 2024 – Present

Walmart Global Tech

Sunnyvale, CA

- Modernized the event and advisory distributed systems architecture — migrated SQL to Cassandra, built Kafka-based event-driven workflows to stream data across downstream systems, and led a GCP database migration across MariaDB, MongoDB, and Spanner — scaling to 15K writes/sec while cutting latency by over 40%.
- Own CI/CD pipelines for backend services and data-pipeline deployments (Kafka streaming, ETL jobs) across specialized data stores including MariaDB, MongoDB, Cassandra, and Spanner.
- Deployed fraud detection models and dynamic rules into production Java services with JavaScript/Python rule-execution engines.
- Led GraalVM-based modernization of the fraud platform's scripting engine, improving scalability and maintainability of rule execution.
- Served as technical lead for returns-fraud and contact-risk initiatives, driving delivery from design through production rollout.
- Core contributor to Walmart's real-time fraud detection platform, which processes hundreds of millions of transactions daily and prevents ~\$2.5B in annual fraud losses.

Senior Software Engineer

Oct 2023 – Jan 2024

Medtronic

Lafayette, CO

- Designed and debugged C++ components for robot-assisted navigation systems used in spinal and cranial surgery procedures.
- Contributed to design, implementation, and reliability testing in a safety-critical medical-device engineering environment.

Research Assistant

Jan 2022 – Sep 2023

Purdue University – National Air and Space Intelligence Center

West Lafayette, IN

- Built a .NET/Python digital forensics platform to mine terabyte-scale disk images via containerized microservices.
- Implemented a Kafka-based analytics pipeline that boosted resource utilization 10x, integrating a React frontend via REST APIs.
- Built a Neo4j knowledge graph modeling file/artifact relationships, cutting investigation time by ~30%.
- Designed an invertible-neural-network RL policy that outperformed DDPG by 138% on OpenAI Gym benchmarks.

Software Engineer II

Jul 2018 – Aug 2021

Adobe

Noida, India

- Built core C++/JavaScript/.NET/Python components for Adobe's extensibility platform, enabling 300 native plugins across internal and external teams.
- Revamped 10 core plugins and platform APIs, cutting application launch time by 50% across flagship Adobe products.
- Led development of 20+ reusable React Native components, cutting feature-development effort 3x; owned Illustrator's sharing feature end-to-end in 6 months.
- Earned a Special Contribution Award for prototyping a Photoshop dialog module and resolving deployment blockers pre-release.

EDUCATION

Master of Science in Computer Science, Machine Learning Specialization

Purdue University

Aug 2021 – Aug 2023

West Lafayette, IN

Teaching Assistant for Statistical Machine Learning and Compilers.

B.Tech, Electronics & Communication Engineering (Minor: CS)

Indian Institute of Technology Roorkee

Jul 2014 – May 2018

Roorkee, India

SELECTED RESEARCH & IP

- Gupta, V. et al. "MANER: Multi-Agent Neural Rearrangement Planning of Objects in Cluttered Environments." IEEE Robotics and Automation Letters.
- Co-inventor, U.S. Patent 10,650,094 — "Predicting Style Breaches Within Textual Content," based on deep learning and NLP research at Adobe.